

3. Row 2 Column 1 = Row 3 Column 1 = Row 4 Column 1 = 0

Row 3 Column 2 = Row 4 Column 2 = 0

Row 4 Column 3 = 0

∴ The condition that for each ~~row~~ ^{having a leading entry} column, the elements below the ~~pivot~~ ^{leading entry} position must be all 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 3 = Row 2 Column 3 = 0

Row 1 Column 2 = 0

∴ The condition that for each column having a leading entry, the elements above the leading entry position must be all 0 is True.

∴ The matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ is in reduced echelon form.

Row 4 has no pivots, since it has no leading entries. ∴ By Theorem A, since every row of A does not have a pivot, the matrix equation $Ax=b$ does not have a solution for all b . So, all arbitrary vectors b are not linear combinations of v_1, v_2 and v_3 (From Thm 4).

∴ $\{v_1, v_2 \text{ and } v_3\}$ do not span $\mathbb{R}^4$. $\square$

In Exercises 23 and 24, mark each statement True or False. Justify each answer.

23. a) The equation $Ax=b$ is referred to as a vector equation.

Solution: On Page 36, it is mentioned that

Equation (3) has the form $Ax=b$. Such an equation is called a matrix equation.

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \quad (3)$$

∴ The given statement is False. $\square$

b) A vector b is a linear combination of the columns of a matrix A if and only if the equation, $Ax=b$ has at least one solution.

Solution: On Page 37, in the topmost grey box, the following fact is mentioned: The equation $Ax=b$ has a solution if and only if b is a linear combination of the columns of A . The given statement is True. $\square$